

Title:

(Re-)Imag(in)ing LOB in Jiangsu Province Electricity Market

Abstract:

The scholars argue image of price trend (open-high-low-close) is useful for daily-level return prediction. In market microeconomic structure, both academic and industry hold consensus that limit order book (LOB) has strong impact on second-level return prediction, but no one provides a theoretical and practical framework to use its information. We firstly transform the LOB data into pixel graphics and further use CNN technique to predict the trend of future return, as opposed to traditional variables extracted from LOB like order imbalance. We rely on second-level LOB charts from electricity market to perform whole analysis. Such LOB-chart based forecasting differs significantly from traditional trend strategies, yields more accurate return predictions, and enables more profitable investment strategies.

Authors:

Jingtao Xu^a, Mei Zhong^{a*}

Jingtao Xu Email: virgil123mry@gmail.com

Mei Zhong Email: zhongm@swfu.edu.cn

Affiliation:

^a College of Economics and Management, Southwest Forestry University,
No. 300 Bailong Temple, Panlong District, Kunming 650224, Yunnan, China

*Corresponding Author:

Mei Zhong

Email: zhongm@swfu.edu.cn

CRedit authorship contribution statement:

Jingtao Xu: Conceptualization, Methodology, Formal analysis, Data curation, Writing – original draft, Writing – review & editing.

Mei Zhong: Supervision, Writing – review & editing.

Declaration of competing interest:

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

(Re-)Imag(in)ing LOB in Jiangsu Province Electricity Market

ARTICLE INFO

Keywords:

Limit order books

Machine learning

Forecasting

ABSTRACT

The scholars argue image of price trend (open-high-low-close) is useful for daily-level return prediction. In market microeconomic structure, both academic and industry hold consensus that limit order book (LOB) has strong impact on second-level return prediction, but no one provides a theoretical and practical framework to use its information. We firstly transform the LOB data into pixel graphics and further use CNN technique to predict the trend of future return, as opposed to traditional variables extracted from LOB like order imbalance. We rely on second-level LOB charts from electricity market to perform whole analysis. Such LOB-chart based forecasting differs significantly from traditional trend strategies, yields more accurate return predictions, and enables more profitable investment strategies.

1. Introduction

Since the 1990s, financial markets have experienced a fundamental transformation in their trading mechanisms. Earlier decades were dominated by floor-based open outcry systems and later by quote-driven structures, both of which have now largely been replaced by electronic, order-driven limit order books (LOBs). This evolution reflects the broader shift in market microstructure toward greater transparency, efficiency, and automation. The LOB functions through two fundamental types of orders: limit orders and market orders (Hollifield, Miller and Sandås, 2004). A limit order represents a trader's commitment, at a given moment, to buy or sell a specific quantity of a security at a predetermined price. By contrast, a market order is executed instantly at the most favorable price available. Limit orders that remain unfilled are retained in the book until they are either executed or withdrawn. Within this microstructure, liquidity is provided by limit orders, while market orders serve to consume that liquidity (Bloomfield, O'hara and Saar, 2005).

LOBs have become central to modern financial markets owing to their distinct advantages. They reduce trading costs by narrowing spreads and lowering fees, while their electronic design allows remote access to a wide pool of participants. By providing both pre-trade and post-trade transparency, LOBs enhance information flow and support efficient price discovery through the aggregation of diverse trading signals. Moreover, they reduce information asymmetry and foster competition, thereby strengthening overall market quality and efficiency.

Recently, financial markets have experienced profound technological transformations, marked by the rapid rise of algorithmic and data-driven trading. Alongside these developments, the growing availability of high-frequency and high-dimensional market data, coupled with advances in computational power, has created fertile ground for the application of deep learning models (Chen, Chen, Huang, Huang and Chen, 2016; Hu, Hu, Yang, Yu, Sung, Zhang, Xie, Liu, Robertson and Hospedales, 2018; Tsantekidis, Passalis, Tefas, Kanninen, Gabbouj and Iosifidis, 2017; Chen, Pelger and Zhu, 2024).

Despite the considerable progress in high-frequency trading research and deep learning applications, existing literature remains heavily concentrated on liquid financial assets such as equities or foreign exchange, where depth, liquidity, and trading frequency naturally suit data-driven modeling (Harris, 2013; Carrion, 2013). By contrast, electricity markets, particularly the hybrid intra-month trading mechanism employed in China have received comparatively little scholarly attention. While a few studies leverage LOB information for volatility forecasting or jump prediction (Mäkinen, Kanninen, Gabbouj and Iosifidis, 2019), they are typically limited to static or low-frequency data, thus failing to capture the subtle, second-level dynamics inherent to electricity markets.

Moreover, although deep learning models such as Convolutional Neural Networks (CNNs) have demonstrated success in stock return prediction using image-transformed price data, these applications are largely limited to benchmark financial markets with regular trading hours, dense order flows, and minimal exogenous intervention. In the case of electricity, however, the non-storability of the asset, sparse order books, and administrative price controls introduce market frictions and information asymmetries that render traditional financial models inadequate. Few, if

*Corresponding author

✉ (); ()

ORCID(s):

any, studies rigorously examine whether deep learning can extract actionable signals from such structurally illiquid, policy-constrained environments.

Lastly, while sentiment-augmented and order-imbalance-based strategies have been explored (Biais, Foucault and Moinas, 2015; Zhang, Zhang, Wang, Yao, Fang and Yu, 2018), these approaches often neglect the microstructural complexities unique to electricity LOBs, such as non-persistent depth, fragmented trading interest, and volatile demand shocks. This absence points to an urgent methodological and empirical gap: a need to develop specialized, high-resolution prediction frameworks tailored to the distinctive attributes of electricity trading.

Our empirical analysis focuses on short-term return prediction in the Jiangsu Province electricity market from February to May 2025. We train a CNN to classify the direction (up or down) of future short-term returns. Specifically, we estimate a single CNN model with a fixed architecture and shared parameters across all training samples, which generates directional forecasts based on LOB images. The input images are constructed from the top five bid and ask prices levels, together with their corresponding volumes, over a fixed 5-second historical window. Based on these images, the model is trained to predict the directional movement of the mid-price at multiple forecast horizons (5, 10, 15, 30 and 60 seconds).

This study contributes to the literature in several important respects. First, while existing studies acknowledge the predictive relevance of the LOB for high-frequency price dynamics, no prior work has proposed a systematic framework that applies CNNs to exploit LOB information for second-level return prediction. Second, rather than relying on traditional handcrafted features, such as order imbalance or aggregated liquidity measures, we introduce a novel image-based representation of the LOB, which allows CNNs to directly capture the geometric and relational patterns embedded in order flow. Third, our empirical results demonstrate that this approach delivers superior predictive accuracy and enhanced trading profitability relative to conventional strategies, underscoring the methodological and practical advantages of treating the LOB as a visual structure rather than a set of pre-defined variables.

Our main empirical finding is that image-based CNN predictions provide powerful forecasts of second-level returns, generating economically meaningful trading profits in a high-frequency market setting. The model exhibits its strongest performance at short horizons, particularly around 10 seconds, where it achieves both high profitability and favorable risk-adjusted outcomes. More broadly, across horizons of 5 to 30 seconds, CNN-based strategies consistently outperform classical technical rules, underscoring their ability to exploit transient and non-linear patterns in limit order book dynamics. To evaluate the economic value of our model, we simulate trading based on predicted class probabilities and benchmark the CNN's performance against three widely used rule-based strategies: momentum, reversal, and trend-following. Of these, momentum strategies remain largely ineffective at short horizons, reversal strategies display intermittent but inconsistent success, and trend-following strategies perform poorly across nearly all settings. Robustness tests further confirm that the CNN model retains strong predictive power in volatile periods, though its advantage diminishes at longer horizons. This pattern supports the view that the model captures genuine patterns in LOB dynamics rather than overfitting to past noise. Taken together, these findings underscore the potential of deep learning methods to extract high-frequency trading signals from market images, offering a powerful complement to conventional rule-based approaches in short-term prediction and execution.

The remainder of the paper is organized as follows. Section 2 reviews the prior literature on high-frequency trading, limit order book and price discovery, and CNNs. Section 3 presents the intuition and detailed mechanics of our CNN framework. We describe the process of converting LOB data into images in Section 4. In Section 5, we report our main empirical analysis. Section 6, we report robustness tests of our main empirical analysis. Section 7 concludes.

2. Literature Review

2.1. High-frequency Trading

High-frequency trading (HFT) influences financial markets in many ways. For example, HFT reduces the bid-ask spread (Harris, 2013), increases price efficiency (Carrion, 2013) and displayed depth in the LOB, and decreases short-term volatility (Hasbrouck and Saar, 2013). In general, HFT plays a dominant part in providing liquidity (Hagströmer and Nordén, 2013; Conrad, Wahal and Xiang, 2015), but probably also withdraw this liquidity during a period of large and temporary selling pressure (Kirilenko, Kyle, Samadi and Tuzun, 2017; Korajczyk and Murphy, 2019). Meanwhile, other papers show that HFT can predict future order flow from other traders and future price movements (Biais et al., 2015; Hoffmann, 2014; Foucault, Hombert and Roşu, 2016; Hirschey, 2021).

HFT does not represent a single trading style but rather a method characterized by extremely short holding periods, rapid order submission, and algorithmic execution. Within this method, HFT strategies are commonly divided into

two broad categories: market-making HFTs, which continuously post limit orders to capture the bid-ask spread while managing inventory risk, and directional HFTs, which seek to forecast short-term price movements and typically rely on aggressive orders. In the case of directional HFTs, order placement strategies based on order book imbalances have been shown to be particularly effective in volatile markets (Brogaard, Hendershott and Riordan, 2019). Aggressive orders in this context exhibit several important properties. First, they are strongly predictive of very short-term price changes and contribute substantially to overall trading volume, accounting for 25%-42% depending on market capitalization (Brogaard, Hendershott and Riordan, 2014). Second, they are closely linked to information flows, responding to market-wide returns, quote updates, macroeconomic announcements, E-mini price changes, and even real-time newswire items (Hirschey, 2021; Kelley and Tetlock, 2013). These findings suggest that directional HFTs frequently trade in anticipation of upcoming information releases.

Aggressive orders, however, come with a significant premium for immediacy. While passive orders tend to have a negative price impact, aggressive orders display strong persistence once executed (Griffiths, Smith, Turnbull and White, 2000). They are usually submitted when spreads and depths are relatively low, and their impact induces a persistent widening of bid-ask spreads and changes in depth. Although market conditions gradually revert, this process may take up to 20 best limit updates following the initial shock (Degryse, Jong and Ravenswaaij, 2005).

Beyond order-level dynamics, a growing body of research emphasizes the role of human behavior and sentiment in shaping price movements (Barberis, 2018). For example, Zhang et al. (2018) propose a predictive model integrating events, sentiment indicators, and quantitative features extracted from diverse sources such as news articles, social media and market data feeds. Earlier studies, however, suggest that while aggregate sentiment measures exhibit strong contemporaneous linkages with market returns, they provide little evidence of short-term predictive power (Brown and Cliff, 2004). This contrast highlights the need for more refined sentiment measures and advanced modeling techniques. Nowadays, the question is no longer whether investor sentiment affects stock prices, but how to measure investor sentiment and quantify its effects. Baker and Wurgler (2007) take the origin of investor sentiment as exogenous and instead focus on its empirical effects.

2.2. Limit Order Book and Price Discovery

Academic evidence suggests that the importance of market design and trading protocols in the determination of agents' decisions to provide liquidity such as price efficiency, volatility, and trading costs (Madhavan, 2000). For example, Large (2009) finds that a LOB induces in expectation equal numbers of market orders and uncanceled limit orders over time in a minimal stationarity property of microstructure dynamics.

The important parameters indicating the shape of order book include the depth, height, slope, resilience and order imbalance (Cao, Hansch and Wang, 2009; Næs and Skjeltorp, 2006). The current state of a LOB substantially affects the price formation and short-term price volatility (Ahn, Bae and Chan, 2001; Pascual and Veredas, 2010). For instance, a greater depth around the best quotes reflects a consensus on the true price (Goettler, Parlour and Rajan, 2005). In contrast, a high preponderance of limit orders away from the best quotes. This mispricing may lead to increasing short-term price volatility and may improve the execution probability of limit orders. Therefore, the shape of order book has significant implications for market efficiency. The resilience of the order book to order flow shocks is one of the important criteria to measure market efficiency, which remains less explored (Degryse et al., 2005; Large, 2007). We infer that this is because the computation of this measure is data intensive and requires intraday high-frequency order book (e.g., order book snapshots at 1-second, 5-second frequencies).

LOB is a channel for volatility information (Foucault, Moinas and Theissen, 2007). The use of LOB data is found to improve the performance of the proposed model in jump prediction (Mäkinen et al., 2019). Sandås (2001) find that break-even conditions for marginal orders in the LOB define the slope of the price schedule offered at a given point in time. After trading halts, A large proportion of the LOB at the reopen is composed of orders submitted during the halt, and that the market-clearing price at the reopen is a good predictor of future prices (Corwin and Lipson, 2000). Informed traders use limit orders when volatility is low (Goettler, Parlour and Rajan, 2009) and realized signals are close to their expected value.

2.3. CNNs

Financial data has to be taken into machine learning methods in asset pricing because of its time dimension (Chen et al., 2024). Gu, Kelly and Xiu (2020) also demonstrate the potential of machine learning in finance through empirical analysis of asset pricing. Deep neural networks (DNNs), especially CNNs, can learn or extract technical indicators (e.g., Bollinger bands, MACD). As a method to predict price movements of financial markets, technical indicators and

technical analysis has been extensively used by market practitioners. Although the theoretical sensibility and empirical reliability of technical analysis have long been a subject of debate, some studies provide evidence of its forecast ability (Lo and MacKinlay, 1988; Timmermann and Granger, 2004). Some scholars combine textual information with historical market data for stock prediction, showing that such approaches outperform models relying on a single input type. For instance, Vargas, De Lima and Vargas, De Lima and Evsukoff (2017) employ a deep learning framework that combines financial news with technical indicators to predict intraday movements of the Standard & Poor's 500 index, while Akita, Yoshihara, Matsubara and Uehara (2016) use recurrent neural networks to integrate news headlines with numerical stock data. An emerging literature in the field of computer vision about image recognition, which using price plots and CNN-based models to forecast stock returns. Existing studies typically offer methodological overviews accompanied by limited-scale empirical validations. For example, Chen et al. (2016) use not only CNNs but also data visualization methods, for the purpose of transforming stock price data into image data to eliminate noise. Hu et al. (2018) use price plot CNNs to cluster individual stocks.

The standard way of approaching LOB modeling has primarily utilized parametric models to concisely capture order book dynamics or simulation-based flexible agent models (Large, 2009; Goettler et al., 2005). More recently, the econometric techniques from other areas (e.g., computer science and information technology) have been adopted to model LOB (Cont, Stoikov and Talreja, 2010; Toth, Palit, Lillo and Farmer, 2015). For example, machine learning models are widely acknowledged to improve the forecast ability in LOB modeling. Tsantekidis et al. (2017) train a CNN on high frequency LOB data and compare with other methods (e.g., multilayer neural networks and support vector machines), showing that CNNs are better than suited for stock price movements.

3. The CNN Model

This section outlines the CNN model architecture and training process, and discusses the rationale for using image-based representations of time-series market data instead of conventional time-series models.

3.1. A Brief Description of the CNN Architecture

To capture intricate spatial patterns embedded in the LOB dynamics, our model adopts a convolutional neural networks (CNN) architecture trained on image-based representations of historical market snapshots. Each image is a pixel graphic derived from LOB features over a lookback window of five consecutive observations, with pixel intensities normalized to the $[0, 1]$ range. While it is theoretically possible to flatten these images and pass them into a conventional feed-forward neural network, such an approach would be both computationally inefficient and structurally myopic. Fully connected architectures, lacking spatial inductive biases, are highly sensitive to variations in position and scale, and typically suffer from overfitting due to their immense parameter count, particularly problematic in data-limited financial contexts.

CNNs overcome these limitations through localized receptive fields, shared weights, and hierarchical abstraction, enabling efficient learning even with modest datasets. Furthermore, their inherent translation invariance allows them to generalize well to subtle structural shifts in the LOB visual patterns.

The architecture of CNN model is shown in Fig. 1. Our proposed CNN architecture consists of two convolutional blocks, each comprising a 2D convolutional layer (with 32 and 64 filters, respectively), batch normalization, ReLU nonlinearity, and a 2×2 max-pooling operation. These blocks progressively extract increasingly abstract and robust local features from the LOB images, while simultaneously reducing spatial resolution to mitigate overfitting. The output of the final convolutional block is flattened and passed through a fully connected network with 64 hidden units, followed by dropout regularization and a final linear layer producing two logits. During inference, a softmax transformation is applied to these logits to yield class probabilities corresponding to the binary classification task—whether the mid-price will rise or fall over a specified forecast horizon. The probability assigned to class 1 reflects the model's confidence in an upward price movement.

3.2. A Brief Description of the CNN Architecture

Our modeling framework begins by transforming the raw limit order book (LOB) time-series data into structured pixel graphics, effectively re-encoding the multidimensional temporal and depth-wise market information into a two-dimensional matrix format. This transformation enables CNNs to leverage their spatial inductive biases, particularly suited to capturing localized, nonlinear dependencies across both price levels and time steps. These nuances are frequently obscured in traditional time-series models.

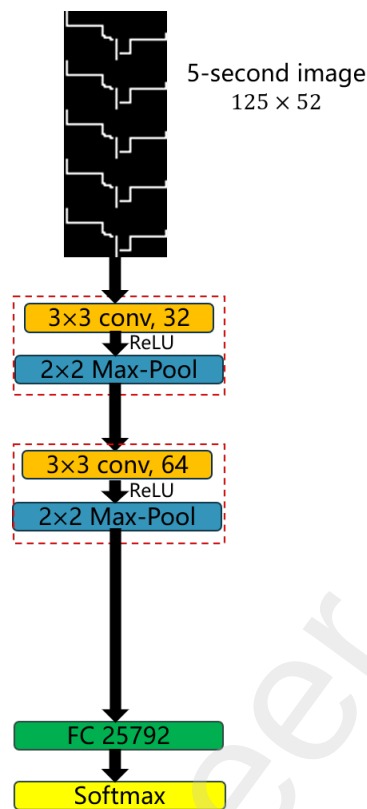


Figure 1: Diagram of the CNN model

By juxtaposing temporal evolution and market depth within a unified image representation, the CNN can discern subtle structural signatures indicative of underlying market regimes. For example, the model becomes capable of distinguishing between stable liquidity configurations, transitory depth distortions, and pronounced bid-ask imbalances. Unlike 1D convolutional approaches that operate over flattened sequences and are limited to capturing sequential dependencies, 2D convolutional filters (e.g., 3×3 kernels) operate over localized spatial patches, allowing the model to detect intricate joint patterns that span both axes such as diagonal spreads of depth shifts or vertically stacked liquidity voids.

Each image encodes multiple key dimensions of the LOB state, including the mid-price, the top five levels of bid and ask quotes, and their corresponding order volumes. These elements are embedded as pixel intensities, forming a visual surface where spatial configurations implicitly reflect directional price pressure, instantaneous liquidity, and market microstructure dynamics. In classical econometric frameworks, modeling such effects would typically require extensive manual feature engineering such as extracting volume-weighted average price (VWAP), order imbalance ratios, or engaging nonlinear modeling paradigms like GARCH or stochastic volatility models. Our approach bypasses these manual steps entirely: through end-to-end training on raw image inputs, the CNN autonomously learns which features are most predictive of short-term price trends.

Fundamentally, encoding LOB states as images allows the model to tap into the same visual pattern-recognition capabilities that underpin much of modern computer vision. Just as a human observer can more readily detect patterns from a heatmap than from a raw numeric table, a CNN can exploit this structured visual representation to discover predictive motifs, which are otherwise difficult to express analytically. This image-based paradigm thus serves not only as a dimensionality-reduction mechanism but as a domain-aware encoding scheme that aligns with the CNN's architectural strengths.

3.3. Training the CNN

Our workflow for model training, validation, and evaluation follows a time-ordered, out-of-sample design. Specifically, we design three sequential training configurations using high-frequency electricity market data from Jiangsu Province, spanning February to May 2025.

In the first configuration, data from February serves as the in-sample set. This dataset is randomly partitioned into training (70%) and validation (30%) subsets. The trained model is then evaluated on March data, representing a genuine out-of-sample environment. The second configuration expands the in-sample window to include both February and March. A stratified 70/30 split is again applied to this combined dataset, and performance is assessed on unseen April data. The third configuration extends this design further: data from February through April is used for model fitting and validation, with May data held out entirely as the final test set. This progressive expansion mirrors a real-world deployment setting, where the model is updated as more historical data becomes available.

We formulate return direction prediction as a classification task. Labels are generated by comparing the mid-price at the current time with the mid-price after a fixed forecast horizon, which we set to one of 5, 10, 15, 30, or 60 seconds depending on the experimental configuration. If the mid-price at the future timestamp exceeds its current value, the label is assigned as $y = 1$ (indicating an upward movement); otherwise, the sample is labeled as $y = 0$. While this binary labeling implicitly includes the "no-change" cases in the negative class, we explicitly address the issue of class imbalance by downsampling the unchanged instances. Specifically, we randomly retain a number of unchanged samples equal to 20% of the total count of directional samples (ups and downs combined), ensuring that the resulting training set is both informative and balanced.

Following this preprocessing, the labeled image samples are divided into training and validation sets using stratified sampling. This methodology reduces bias from prevailing market regimes (e.g., prolonged bullish or bearish trends) and enhances the robustness of the classifier under varying market conditions.

The CNN produces a two-dimensional output vector $z = [z_0, z_1]$, referred to as logits, which correspond to the model's unnormalized confidence scores for the two classes. Model training minimizes the standard cross-entropy loss function:

$$L(y, z) = -\log \left(\frac{\exp(z_y)}{\sum_{i=0}^1 \exp(z_i)} \right) \quad (1)$$

Here, $y \in \{0, 1\}$ is the ground truth label, and the softmax operation is applied internally to transform logits into valid class probabilities. This objective encourages the model to allocate high confidence to the correct class while penalizing erroneous predictions proportionally.

We optimize this loss using the Adam optimizer, with an initial learning rate of 1×10^{-5} , a batch size of 128, and an L_2 weight decay of 1×10^{-4} (Srivastava, Hinton, Krizhevsky, Sutskever and Salakhutdinov, 2014). To prevent overfitting and promote generalization, we implement an early stopping mechanism: training is halted when the validation loss fails to improve for two consecutive epochs.

In the fully connected layers, we apply dropout with a probability of 0.5 to mitigate the co-adaptation of neurons and encourage the learning of robust features. L_2 regularization further constrains model complexity by penalizing excessively large weight magnitudes.

The convolutional backbone of the network incorporates ReLU activations, batch normalization, and 2×2 max-pooling operations. These elements jointly introduce nonlinearity, promote numerical stability during training, and impart spatial invariance to the learned features. Importantly, dropout is not applied within the convolutional layers to preserve the spatial integrity of local feature representations. This architectural design, combined with early stopping and regularization, ensures stable, efficient, and generalizable training performance across diverse market conditions.

4. LOB Data

In this section, we outline the process used to transform historical electricity market data into image-based inputs for the CNN forecasting model. The dataset employed in this study was obtained from trading information by the Jiangsu Power Exchange Center. Specifically, the data were collected through a custom-developed Python web crawler from the Jiangsu Power Exchange Center, encompassing per-second market transaction volumes, prices, and related information.

In Jiangsu Province of China, at each month, Government will open the electricity exchange for several days. At each electricity trading day, the trading hour is fixed between 10:00 am and 11:00 am. So, based on 1-second frequency



Figure 2: BTC/USDT Limit Order Book and Market Depth Chart from Binance

observation, we generally will have about 3600 second-level LOB and transaction result. Our dataset covers all trading days from February to May 2025, amounting to 49 trading days in total, and includes approximately 160,000 second-level LOB and transaction result. Each snapshot captures the full state of the LOB at that second, including prices, volumes, and executed trades, providing a rich basis for modeling market microstructure dynamics and predicting short-term returns.

Fig. 2 illustrates a 1-second snapshot of the Bitcoin (BTC/USDT) depth chart as displayed on a cryptocurrency trading platform. The right panel displays a market depth chart, where the green and red areas respectively represent cumulative buy-side and sell-side liquidity. This visual representation provides insight into short-term market sentiment and order flow, serving as the basis for constructing LOB images used in our model.

Fig. 3 illustrates a black-and-white visual representation of the LOB over a rolling five-second window. Each image is constructed on a fixed canvas of 125 pixels in height and 52 pixels in width. The horizontal axis encodes price levels: the left half is reserved for the top five bid prices, the right half for the top five ask prices, while the vertical line is dynamically positioned between the best bid and the best ask. The vertical axis, in turn, translates the cumulative order volumes into pixel coordinates through a logarithmic transformation and normalization, ensuring that both small and large volumes can be visualized within the same frame. The temporal dimension unfolds from bottom to top: the lowest 25-pixel band corresponds to the first second within the window, while each subsequent band above represents the next second, culminating in the most recent snapshot at the top. Within each band, the bid and ask prices are mapped horizontally according to their relative distance from the mid-price, while the corresponding cumulative volumes are plotted vertically. These coordinates are then connected in a step-line fashion, producing ladder-like curves for both bids and asks. Finally, a vertical line is drawn at the mid-price for each second, visually anchoring the bid-ask dynamics around the evolving price center. In this way, the sequence of images provides an intuitive yet rigorous visualization of order book fluctuations.

This design intricately emphasizes the structural aspects of the order book over time, showcasing the distribution of liquidity across varying price levels and highlighting the symmetry between the buy and sell sides. The visualization captures the dynamic movement of the order book, making it possible to discern subtle trends and patterns in the market depth as it evolves.

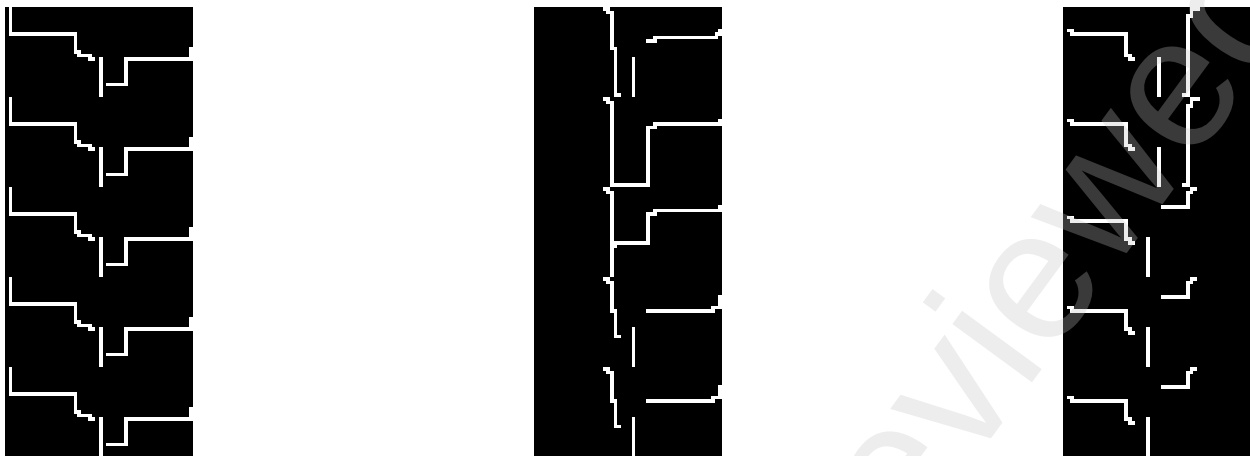


Figure 3: Generated 5-second LOB Image

The decision to focus on the top 5 bid-ask prices and their corresponding volumes, in conjunction with the mid-price, is firmly rooted in both practical considerations and the foundational principles of market microstructure theory. From a market microstructure standpoint, the top levels of the limit order book (typically the first 5 – 10 levels) represent the most immediately executable liquidity, encapsulating the most up-to-date supply and demand conditions. Empirical research (Conrad et al., 2015; Hautsch and Huang, 2012) consistently demonstrates that price movements in high-frequency trading environments are frequently initiated by shifts in these near-best quotes, as they directly mirror the trading intentions of the market's most active participants.

By limiting the model's input to the top 5 levels, we strike an optimal balance between information richness and computational efficiency. Including an excessive number of levels could introduce noise, as the deeper layers of the book tend to become stale, offering less relevant data for short-term predictions. On the other hand, focusing on too few levels risks overlooking critical liquidity shifts and failing to capture the imbalances just beyond the best bid and ask.

Moreover, we incorporate the average of the best bid and best ask prices, which serves as a robust, widely accepted proxy for the true market price. Including this mid-price in each snapshot ensures that the model can effectively normalize price movements, fostering consistency across varying volatility regimes. This compact, yet rich visual representation empowers the CNN to discern both spatial and temporal regularities in the LOB, thereby enhancing the model's ability to forecast short-term returns with greater accuracy.

5. Empirical Result

In this study, we adopt a single input design that encodes LOB data from the past 5 seconds into images. Each image is labeled with a "1" if the mid-price increases over a specified forecast horizon (5, 10, 15, 30, or 60 seconds), and "0" otherwise. For each distinct forecast horizon, we independently train the CNN model five times to account for the inherent stochasticity in the optimization process, such as weight initialization and data shuffling. The resulting predictions are then averaged to enhance stability, following the methodology outlined by Gu et al. (2020).

To optimize computational efficiency, we refrain from utilizing a rolling retraining approach. Instead, each model is trained and validated once using in-sample data from February to April 2025. For model training, 70% of the samples are randomly selected, while the remaining 30% are reserved for validation. Once trained, the model is employed to generate predictions on out-of-sample data from May 2025.

5.1. Short-horizon Model Performance

To assess the overall performance of the proposed model, we report both in-sample and out-of-sample (OOS) McFadden R^2 , which is suitable only for regression tasks involving continuous outcomes, McFadden R^2 provides a more appropriate measure of goodness-of-fit for classification models such as logistic regression and neural networks. The implementation follows three steps. First, we calculate the log-likelihood of the estimated model (LL_{model}) by

applying the softmax output to the true class labels. For each observation, the predicted probability of its actual class is extracted, the logarithm is taken, and the results are summed across all samples. Second, we compute the log-likelihood of the null model (LL_{null}), which assumes equal probability for each class. In the binary setting, this reduces to $LL_{\text{null}} = N \cdot \ln(0.5)$, where N denotes the number of samples. Finally, McFadden R^2 is obtained as:

$$R^2_{\text{McFadden}} = 1 - \frac{LL_{\text{model}}}{LL_{\text{null}}} \quad (2)$$

Table 1 summarizes the model's predictive performance across five forecast horizons using LOB image inputs and illustrates how the effectiveness of short-horizon return direction prediction varies as the prediction window lengthens. In addition, the table includes out-of-sample classification accuracy on May data, offering an evaluation of the model's generalization ability to unseen market conditions.

Table 1
Model performance across different prediction horizons

Horizon	5s	10s	15s	30s	60s
Training set McFadden R^2	0.24	0.39	0.48	0.59	0.70
Validation set McFadden R^2	0.12	0.29	0.38	0.52	0.65
Test set McFadden R^2	0.16	0.15	0.22	0.12	0.02
OOS accuracy	0.71	0.69	0.72	0.69	0.71

A clear upward trend is observed in the training set's McFadden R^2 , rising steadily from 0.2377 at the 5-second horizon to 0.7048 at 60 seconds. This progression indicates that the model captures increasingly rich predictive structure as the forecast horizon extends. Intuitively, shorter-term price movements are often more erratic and driven by noise, rendering them harder to predict. In contrast, longer horizons tend to reveal more pronounced patterns, such as autocorrelation and trend persistence, which the model is able to learn more effectively.

On the validation set, a similar improvement is observed: McFadden R^2 increases from 0.1225 to 0.6453 as the horizon lengthens. This consistent performance across unseen in-sample data suggests strong generalization capacity. Furthermore, the narrowing gap between training and validation R^2 at extended horizons implies reduced overfitting and greater model robustness in those settings. However, the test set (out-of-sample) McFadden R^2 follows a different trajectory. It peaks at the 15-second horizon with a value of 0.2191 but declines sharply thereafter, reaching just 0.022 at 60 seconds. This reversal suggests either overfitting to in-sample patterns that do not persist into the OOS period, or a shift in the underlying price dynamics during May, which undermines the model's predictive accuracy. Such non-monotonic behavior underscores the fundamental challenge of temporal non-stationarity in high-frequency financial markets—patterns that hold over one time span may degrade or vanish in another, especially as longer horizons introduce additional layers of macroeconomic or structural noise.

Despite this decline in R^2 on the test set, the out-of-sample classification accuracy remains relatively stable across horizons, fluctuating within a narrow range from 0.6924 to 0.7173. This stability suggests that, although the model's confidence in its predictions may weaken, the classification boundary it learns remains reasonably robust for directional inference across varying timeframes.

Table 2 presents the normalized out-of-sample (OOS) confusion matrix for the CNN model's directional forecasts across multiple short-term return horizons. Each cell of the matrix represents the proportion of test set samples that fall into one of four prediction-outcome categories: True Positive (TP), False Negative (FN), False Positive (FP), and True Negative (TN). This matrix serves not only to evaluate the overall classification accuracy but also to assess the model's directional bias. Specifically, it helps identify whether the model exhibits a tendency to overpredict upward trends or, conversely, demonstrates a more conservative approach in predicting downturns.

A clear and striking pattern emerges in the matrix: there is a strong asymmetry between "True Up" and "True Down" samples. Across all horizons, the proportion of "True Down" cases (i.e., mid-price decreases after the prediction point) dominates the test set. For example, "True Down" cases account for 96.6% of all samples at the 5-second horizon. Even at the 60-second horizon, upward trends account for less than 17% of the total test set. This imbalance reflects the inherent dynamics of high-frequency market microstructure, where price movements are typically small, short-lived, and more frequently downward due to factors such as bid-ask bounce, short-term liquidity pressures, or quote revisions.

Table 2
OOS Win Rate Confusion Matrix

Horizon	TP	FN	FP	TN
5s	0.01357	0.02028	0.26215	0.70399
10s	0.02193	0.03070	0.27868	0.66869
15s	0.02728	0.04161	0.23266	0.69845
30s	0.04424	0.06538	0.23813	0.65226
60s	0.04391	0.12413	0.16174	0.67023

(Foucault, Kadan and Kandel, 2005). This inherent skew presents a challenge for classifiers: if the model were to predict only the majority class, its accuracy would artificially increase.

The OOS confusion matrix also reveals a consistent directional bias in the model's predictions, particularly a conservative tendency to favor downward forecasts across all return horizons. This is evident in the high proportions of "True Down – Forecast Down" cases, such as 70.4% at 5 seconds and 67.0% at 60 seconds, paired with relatively low TP rates for upward movements, which stand at only 1.36% at 5 seconds. From a calibration standpoint, this pattern suggests that the model is more confident in predicting trend continuation or short-term reversals toward the downside, and it is less likely to assign high probabilities to upward trends unless there are strong distinguishing features. This asymmetry may arise from the label imbalance in the training set, as well as the limited visual variation among upward patterns in the LOB image data, which hampers the model's ability to generalize upward signals effectively.

As the prediction horizon increases, the model's predictive confidence shows signs of degradation. This is likely due to the increasing noise in future returns and the non-stationary nature of high-frequency market microstructure, both of which reduce the correlation between current order book features and future price movements. Empirically, this is reflected in a rising FN rate, which increases from 2.0% at 5 seconds to 12.4% at 60 seconds. In contrast, the FP rate decreases over the same horizon range, indicating growing conservatism and possibly reflecting lower softmax confidence thresholds as the horizon extends.

Overall, the CNN-based model exhibits consistent performance in forecasting short-term return directions, particularly within the 5 to 15-second window. McFadden R^2 values demonstrate a robust in-sample fit and a moderate degree of generalization. Simultaneously, the out-of-sample classification accuracy remains stable across the various horizons, while the confusion matrix analysis highlights a directional conservatism that mirrors the class imbalance inherent in the LOB data.

To ascertain whether these results are consistent across different periods and not limited to the May dataset, we conduct a series of robustness tests utilizing alternative OOS periods. The additional findings from these tests are presented in Section 6.

5.2. Trading

We implement trading strategies by converting both model forecasts and benchmark signals into discrete positions through a rolling quantile procedure. For each valid decision point, the CNN model produces an upward probability, which is mapped into a long or short signal (+1/ − 1) if the value lies in the upper or lower quantile of its rolling distribution. Analogous rules are applied to the momentum, short-term reversal, and trend-following indicators derived from mid-price dynamics. Importantly, all strategies are executed on the same set of decision points, ensuring that performance differences are attributable to the forecasting methods rather than timing advantages.

For each trade initiated at time t , the realized return is defined as

$$r_t = \text{investment} \times \frac{P_{t+H} - P_t}{P_t}. \quad (3)$$

where the investment takes values of ± 1 . Here, P_t denotes the mid-price at time t and P_{t+H} the mid-price after horizon H . Daily returns are then aggregated to compute annualized performance statistics, including return, volatility, Sharpe ratio and the total number of trades.

Table 3 presents the results of applying the CNN-based trend prediction model to out-of-sample data across various short-term return horizons. At very short horizons (5s, 10s, and 15s), the strategy achieves strong and economically meaningful results. For instance, the annualized return reaches 59.5% at the 5-second horizon and 62.82% at 15 seconds,

Table 3
OOS Trading

Horizon	5s	10s	15s	30s	60s
Total trades	7880	6442	6653	4332	3991
Annualized returns	0.5950	0.3605	0.6282	0.6185	-2.2348
Annualized volatility	0.0694	0.0307	0.0911	0.2116	0.2556
Sharpe ratio	8.57	11.74	6.89	2.92	-8.74

while the corresponding Sharpe ratios remain high at 8.57 and 6.89. The 10-second horizon delivers the most favorable risk-adjusted performance, with a Sharpe ratio of 11.74 and relatively low volatility of 3.07%.

At longer horizons, however, the profitability deteriorates. Although the 30-second horizon still yields a positive annualized return of 61.85%, the Sharpe ratio drops to 2.92, reflecting the higher volatility of 21.16%. The degradation becomes even more pronounced at the 60-second horizon, where the strategy generates a large negative annualized return of -223.48% and a Sharpe ratio of -8.74. This pattern suggests that the CNN model is particularly effective in capturing short-term predictive signals from limit order book images, but its forecasts lose reliability as the horizon extends, where noise and mean-reverting microstructure effects dominate price dynamics.

The CNN-based trend prediction model demonstrates notable profitability and exceptional risk-adjusted returns, particularly within high-frequency horizons (5s, 10s, and 15s). These results validate the potential of deep learning models for real time execution in algorithmic trading, especially when complemented by conservative position sizing and stringent transaction cost management. However, as the prediction horizon extends (30s and 60s), both the model's predictive calibration and its economic performance decline, suggesting that CNN models trained on short-term LOB images are less adept at capturing longer-term price movements.

6. Robust Test

While Section 5 demonstrates the model's effectiveness on May OOS data, robust financial modeling requires evidence that its performance is not limited to a single market condition. To address this, we conduct two complementary robustness tests. First, we evaluate the model's predictive power under different volatility regimes to assess its stability across different market environments. Second, we benchmark the CNN model against several classic technical trading strategies to determine whether its superior performance stems from capturing genuine market signals rather than overfitting or data specific patterns.

6.1. Performance Across Market Volatility Periods

To examine whether the CNN model retains its predictive accuracy and trading profitability across different market environments, we conduct a robustness analysis stratified by local realized volatility. Specifically, we compute rolling volatility based on intraday mid-price returns. For each test sample at time index t , the back-looking realized volatility is calculated over a 60-second window as:

$$\sigma_t = \text{std} \left\{ \frac{P_{i+1} - P_i}{P_i} \mid i \in [t - \omega, \dots, t - 1] \right\}, \quad (4)$$

where P_t denotes the mid-price at time t , and $\omega = 60$ seconds is the estimation window. This measure captures the intensity of recent return fluctuations. After computing volatility for all valid test samples, we split them into High-volatility and Low-volatility groups using the median volatility as the threshold, ensuring balanced subsample sizes.

Table 4 and 5 present the model's performance when the out-of-sample test set is stratified by volatility. Several important patterns emerge. In high-volatility periods, the CNN model translates turbulent market conditions into profitable short-term strategies. At horizons of 5 to 30 seconds, annualized returns remain positive, ranging from 23.42% to 34.30%, while Sharpe ratios are consistently above 3. Notably, the 5-second horizon achieves the highest Sharpe ratio of 9.43, indicating that the model is particularly effective in extracting predictive signals from order flow when market activity is intense. However, performance deteriorates substantially at the 60-second horizon, with negative McFadden R^2 , a Sharpe ratio of -6.95, and an annualized return of -158.19%. This suggests that as the forecast horizon lengthens, noise and mean reversion dominate price dynamics, limiting profitability. In contrast,

during low-volatility periods, model maintains robust classification accuracy, with McFadden R^2 values as high as 0.4451 at the 15-second horizon. Trading performance is also favorable at short to medium horizons. For example, at 5 and 15 seconds, Sharpe ratios reach 12.56 and 7.98, respectively, while annualized returns are 18.05% and 28.52%. Interestingly, the 30-second horizon yields the strongest profitability in quiet markets, with an annualized return of 95.13% and a Sharpe ratio of 7.96. Yet, as in the high-volatility case, performance collapses at the 60-second horizon, where both Sharpe ratios and returns turn sharply negative.

Table 4
High-volatility Trading

Horizon	McFadden R^2	Sharpe Ratio	Annualized Returns
5s	0.08	9.43	0.3317
10s	0.02	5.51	0.2342
15s	0.26	3.67	0.3430
30s	0.15	3.22	0.2889
60s	-0.02	-6.95	-1.5819

Table 5
Low-volatility Trading

Horizon	McFadden R^2	Sharpe Ratio	Annualized Returns
5s	0.19	12.56	0.1805
10s	0.23	5.63	0.1263
15s	0.45	7.98	0.2852
30s	0.35	7.96	0.9513
60s	0.20	-3.90	-0.6529

These results reveal a clear asymmetry in the CNN model's performance across volatility regimes. In high-volatility environments, the model excels at extracting predictive signals from order flow, yielding consistently high Sharpe ratios and positive returns at short horizons, though profitability deteriorates once the forecast horizon extends to 60 seconds. In contrast, in low-volatility periods the model continues to exhibit strong classification accuracy and generates attractive returns at shorter and medium horizons, yet its profitability collapses at longer horizons due to the absence of strong directional movements and the dominance of noise. These findings confirm that the CNN framework is particularly well suited for high-frequency trading under volatile market conditions, while its effectiveness diminishes in calmer environments, especially when the prediction horizon is lengthened.

6.2. Comparison with Traditional Strategies

Having established the CNN model's differential performance across volatility periods, We benchmark it against three standard rule-based technical strategies: (i) a short-term reversal strategy, which buys after negative returns over the past 10 seconds and sells after positive returns, thereby exploiting potential price reversals; (ii) a momentum strategy, which evaluates cumulative returns from 60 to 10 seconds prior, going long if the cumulative return is positive and short otherwise; (iii) a trend-following strategy, which compares the current price with its moving average over a window equal to the prediction horizon, going long when the price exceeds the moving average and short otherwise. These strategies are standard approaches commonly employed in quantitative finance.

For each strategy, we evaluate both the annualized return and annualized volatility across five prediction horizons: 5s, 10s, 15s, 30s, and 60s. The comparative results are presented in Fig. 4.

Our findings demonstrate that the CNN model achieves consistently superior performance relative to the benchmark strategies, particularly at shorter horizons. The model performs best at the 10-second horizon, where it records an annualized return of 36.05% with a volatility of only 3.07%, resulting in the highest Sharpe ratio among all strategies and horizons. Similar robustness is observed at the 5-second and 15-second horizons, with annualized returns of 59.5% and 62.8%, respectively, accompanied by Sharpe ratios above 6. In contrast, the performance deteriorates sharply at longer horizons, most notably at 60 seconds, where the CNN model produces large negative returns (-223.5%) and a strongly negative Sharpe ratio (-8.74), highlighting its limitation in capturing more persistent dynamics.

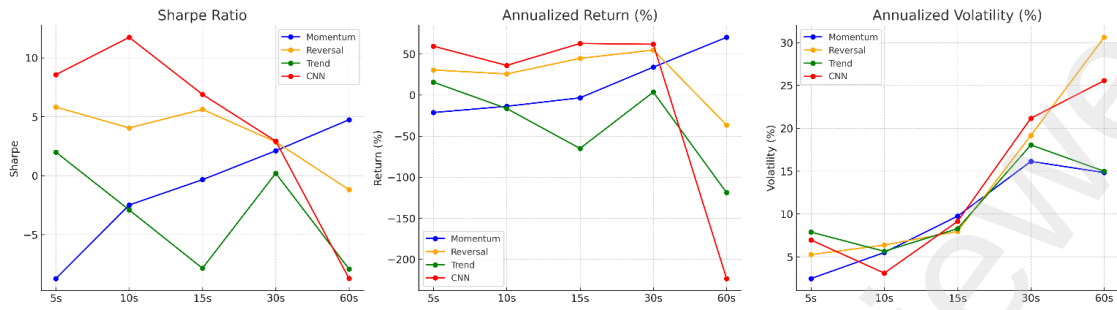


Figure 4: Trading by Traditional Technical Strategies

The momentum strategy exhibits weak and generally negative performance at shorter horizons, with Sharpe ratios ranging from -8.75 at 5 seconds to -2.50 at 10 seconds. It only becomes profitable at longer horizons, delivering its strongest outcome at 60 seconds with an annualized return of 70.3% and a Sharpe ratio of 4.74. The reversal strategy yields relatively strong returns at 5, 10, and 15 seconds (30.5%, 25.7%, and 44.7%, respectively), with corresponding Sharpe ratios ranging from 4 to 6. However, its performance is inconsistent, collapsing at the 60-second horizon with negative returns (-36.6%) and high volatility (30.6%). Finally, the trend-following strategy performs poorly across most horizons, producing negative returns at 10, 15, and 60 seconds, with only marginally positive results at 5 and 30 seconds, confirming its limited applicability in high-frequency environments.

The trend-following strategy assumes that prices move in prolonged directional patterns relative to moving averages. However, in high-frequency markets, such trends are rare. Short-term price movements are typically fragmented and shaped more by limit order placement, cancellation, and matching mechanics rather than by macroeconomic or behavioral drivers. Furthermore, in our test set, we observed that the majority of price movements were stagnant. Due to the binary labeling scheme employed in this study, observations with no price change over the prediction horizon were assigned to the “down” class. This situation causes the trend-following strategy, which depends on identifying momentum above a moving average, to frequently generate false-positive long signals in a market that is either flat or declining. As these conditions do not align with sustained trends, the strategy struggles to establish effective directional momentum and incurs consistent losses.

In summary, these rule-based strategies are ill-suited to the dynamics of high-frequency markets, where non-linear, rapidly shifting, and low-signal-to-noise patterns dominate. Their rigid signal structures lack the adaptability needed to interpret the subtle variations in limit order book states. In contrast, the CNN model can learn these complex, context-dependent patterns directly from LOB image inputs, making it far more effective for short-term prediction and trading.

7. Conclusion

This paper contributes to the growing body of literature at the intersection of computer vision and financial modeling by applying a Convolutional Neural Network (CNN) to predict second-level returns using Limit Order Book (LOB) image data. By transforming raw market microstructure information into visual representations, the CNN is able to detect complex and non-linear patterns that traditional rule-based methods often fail to capture. Across multiple short-term forecast horizons, our empirical findings demonstrate that the image-based model delivers superior return predictability and trading performance, particularly within the 5–30 second range.

We further validate the model’s robustness through tests across different volatility regimes and through systematic comparisons with classical technical strategies such as momentum, reversal, and trend-following. While rule-based approaches exhibit horizon-dependent weaknesses—momentum strategies performing poorly at short horizons but gaining strength at longer horizons, reversal strategies showing intermittent success yet lacking consistency, and trend-following strategies remaining largely ineffective—the CNN model consistently dominates in the short-run. However, at longer horizons such as 60 seconds, its performance deteriorates sharply, highlighting the temporal limits of its predictive power.

One of the most important implications of our findings is the model’s ability to capture subtle, high-frequency patterns embedded in LOB structures—patterns that may reflect transient liquidity imbalances, order flow pressure, or

fleeting arbitrage opportunities. Unlike conventional time-series methods, the CNN learns from spatial configurations of market data, offering a fundamentally different lens through which to interpret market dynamics. In doing so, it not only serves as a predictive tool but also points toward a richer modeling paradigm: one that aligns more closely with how human traders visually interpret order books and respond to their structure in real time.

The study represents a modest but promising step toward bridging the gap between machine perception and high-frequency financial decision-making. Future research could explore extending this framework to portfolio-level applications, incorporating multi-asset visual inputs, or integrating attention-based architectures to enhance interpretability. As markets continue to evolve in complexity and speed, image-based approaches provide a novel and powerful methodology for decoding the rapidly shifting signals embedded within market microstructure.

CRedit authorship contribution statement

... .

References

- Ahn, H.J., Bae, K.H., Chan, K., 2001. Limit orders, depth, and volatility: Evidence from the stock exchange of hong kong. *J. Finance* 56, 767–788.
- Akita, R., Yoshihara, A., Matsubara, T., Uehara, K., 2016. Deep learning for stock prediction using numerical and textual information, in: *Proc. 2016 IEEE/ACIS 15th Int. Conf. Comput. Inf. Sci. (ICIS)*, pp. 1–6.
- Baker, M., Wurgler, J., 2007. Investor sentiment in the stock market. *J. Econ. Perspect.* 21, 129–151.
- Barberis, N., 2018. Psychology-based models of asset prices and trading volume, in: *Handbook of Behavioral Economics: Applications and Foundations* 1. Elsevier, Amsterdam. volume 1, pp. 79–175.
- Biais, B., Foucault, T., Moinas, S., 2015. Equilibrium fast trading. *J. Financ. Econ.* 116, 292–313.
- Bloomfield, R., O'hara, M., Saar, G., 2005. The “make or take” decision in an electronic market: Evidence on the evolution of liquidity. *J. Financ. Econ.* 75, 165–199.
- Brogaard, J., Hendershott, T., Riordan, R., 2014. High-frequency trading and price discovery. *Rev. Financ. Stud.* 27, 2267–2306.
- Brogaard, J., Hendershott, T., Riordan, R., 2019. Price discovery without trading: Evidence from limit orders. *J. Finance* 74, 1621–1658.
- Brown, G.W., Cliff, M.T., 2004. Investor sentiment and the near-term stock market. *J. Empir. Finance* 11, 1–27.
- Cao, C., Hansch, O., Wang, X., 2009. The information content of an open limit-order book. *J. Futures Mark.* 29, 16–41.
- Carrion, A., 2013. Very fast money: High-frequency trading on the nasdaq. *J. Financ. Mark.* 16, 680–711.
- Chen, J.F., Chen, W.L., Huang, C.P., Huang, S.H., Chen, A.P., 2016. Financial time-series data analysis using deep convolutional neural networks, in: *Proc. 2016 7th Int. Conf. Cloud Comput. Big Data (CCBD)*, pp. 87–92.
- Chen, L., Pelger, M., Zhu, J., 2024. Deep learning in asset pricing. *Manag. Sci.* 70, 714–750.
- Conrad, J., Wahal, S., Xiang, J., 2015. High-frequency quoting, trading, and the efficiency of prices. *J. Financ. Econ.* 116, 271–291.
- Cont, R., Stoikov, S., Talreja, R., 2010. A stochastic model for order book dynamics. *Oper. Res.* 58, 549–563.
- Corwin, S.A., Lipson, M.L., 2000. Order flow and liquidity around nyse trading halts. *J. Finance* 55, 1771–1801.
- Degryse, H., Jong, F.D., Ravenswaaij, M.V., 2005. Aggressive orders and the resiliency of a limit order market. *Rev. Finance* 9, 201–242.
- Foucault, T., Hombert, J., Roşu, I., 2016. News trading and speed. *J. Finance* 71, 335–382.
- Foucault, T., Kadan, O., Kandel, E., 2005. Limit order book as a market for liquidity. *Rev. Financ. Stud.* 18, 1171–1217.
- Foucault, T., Moinas, S., Theissen, E., 2007. Does anonymity matter in electronic limit order markets? *Rev. Financ. Stud.* 20, 1707–1747.
- Goettler, R.L., Parlour, C.A., Rajan, U., 2005. Equilibrium in a dynamic limit order market. *J. Finance* 60, 2149–2192.
- Goettler, R.L., Parlour, C.A., Rajan, U., 2009. Informed traders and limit order markets. *J. Financ. Econ.* 93, 67–87.
- Griffiths, M.D., Smith, B.F., Turnbull, D.A.S., White, R.W., 2000. The costs and determinants of order aggressiveness. *J. Financ. Econ.* 56, 65–88.
- Gu, S., Kelly, B., Xiu, D., 2020. Empirical asset pricing via machine learning. *Rev. Financ. Stud.* 33, 2223–2273.
- Hagströmer, B., Nordén, L., 2013. The diversity of high-frequency traders. *J. Financ. Mark.* 16, 741–770.
- Harris, L., 2013. What to do about high-frequency trading. *Financ. Anal. J.* 69, 6–9.
- Hasbrouck, J., Saar, G., 2013. Low-latency trading. *J. Financ. Mark.* 16, 646–679.
- Hautsch, N., Huang, R., 2012. The market impact of a limit order. *J. Econ. Dyn. Control* 36, 501–522.
- Hirschey, N., 2021. Do high-frequency traders anticipate buying and selling pressure? *Manag. Sci.* 67, 3321–3345.
- Hoffmann, P., 2014. A dynamic limit order market with fast and slow traders. *J. Financ. Econ.* 113, 156–169.
- Hollifield, B., Miller, R.A., Sandås, P., 2004. Empirical analysis of limit order markets. *Rev. Econ. Stud.* 71, 1027–1063.
- Hu, G., Hu, Y., Yang, K., Yu, Z., Sung, F., Zhang, Z., Xie, F., Liu, J., Robertson, N., Hospedales, T., e.a., 2018. Deep stock representation learning: From candlestick charts to investment decisions, in: *Proc. 2018 IEEE Int. Conf. Acoust. Speech Signal Process. (ICASSP)*, pp. 2706–2710.
- Kelley, E.K., Tetlock, P.C., 2013. How wise are crowds? insights from retail orders and stock returns. *J. Finance* 68, 1229–1265.
- Kirilenko, A., Kyle, A.S., Samadi, M., Tuzun, T., 2017. The flash crash: High-frequency trading in an electronic market. *J. Finance* 72, 967–998.
- Korajczyk, R.A., Murphy, D., 2019. High-frequency market making to large institutional trades. *Rev. Financ. Stud.* 32, 1034–1067.
- Large, J., 2007. Measuring the resiliency of an electronic limit order book. *J. Financ. Mark.* 10, 1–25.
- Large, J., 2009. A market-clearing role for inefficiency on a limit order book. *J. Financ. Econ.* 91, 102–117.
- Lo, A.W., MacKinlay, A.C., 1988. Stock market prices do not follow random walks: Evidence from a simple specification test. *Rev. Financ. Stud.* 1, 41–66.
- Madhavan, A., 2000. Market microstructure: A survey. *J. Financ. Mark.* 3, 205–258.

- Mäkinen, Y., Kanninen, J., Gabbouj, M., Iosifidis, A., 2019. Forecasting jump arrivals in stock prices: new attention-based network architecture using limit order book data. *Quant. Finance* 19, 2033–2050.
- Næs, R., Skjeltorp, J.A., 2006. Order book characteristics and the volume–volatility relation: Empirical evidence from a limit order market. *J. Financ. Mark.* 9, 408–432.
- Pascual, R., Veredas, D., 2010. Does the open limit order book matter in explaining informational volatility? *J. Financ. Econometr.* 8, 57–87.
- Sandås, P., 2001. Adverse selection and competitive market making: Empirical evidence from a limit order market. *Rev. Financ. Stud.* 14, 705–734.
- Srivastava, N., Hinton, G., Krizhevsky, A., Sutskever, I., Salakhutdinov, R., 2014. Dropout: A simple way to prevent neural networks from overfitting. *J. Mach. Learn. Res.* 15, 1929–1958.
- Timmermann, A., Granger, C.W.J., 2004. Efficient market hypothesis and forecasting. *Int. J. Forecast.* 20, 15–27.
- Toth, B., Palit, I., Lillo, F., Farmer, J.D., 2015. Why is equity order flow so persistent? *J. Econ. Dyn. Control* 51, 218–239.
- Tsantekidis, A., Passalis, N., Tefas, A., Kanninen, J., Gabbouj, M., Iosifidis, A., 2017. Forecasting stock prices from the limit order book using convolutional neural networks, in: *Proc. 2017 IEEE 19th Conf. Business Informatics (CBI)*, Vol.1, pp. 7–12.
- Vargas, M.R., De Lima, B.S.L.P., Evsukoff, A.G., 2017. Deep learning for stock market prediction from financial news articles, in: *Proc. 2017 IEEE Int. Conf. Comput. Intell. Virtual Environ. Meas. Syst. Appl. (CIVEMSA)*, pp. 60–65.
- Zhang, X., Zhang, Y., Wang, S., Yao, Y., Fang, B., Yu, P.S., 2018. Improving stock market prediction via heterogeneous information fusion. *Knowl.-Based Syst.* 143, 236–247.